
HOMEWORK DAY 17 – *Derivatives and the shape of the graph §3.3*

1. §3.3: 1

2. §3.3: 8

3. §3.3: 10

4. For the function $f(x) = 3x^4 - 4x^3 + 2$

(a) Find the intervals where the function is increasing/decreasing and the coordinates of all local max/min.

(b) Find the intervals where the function is concave up/down and the coordinates at all inflection points.

(c) Use the above information to sketch a rough graph of the function.

5. For the function $f(x) = \cos^2 x - 2 \sin x$, $x \in [0, 2\pi]$

(a) Find the intervals where the function is increasing/decreasing and the coordinates of all local max/min.

(b) Find the intervals where the function is concave up/down and the coordinates at all inflection points.

(c) Use the above information to sketch a rough graph of the function.

6. §3.4: 4

7. Find the following limits. Carefully show your work.

(a) §3.4: 9

(b) §3.4: 11

(c) §3.4: 12

8. For the function $f(x) = \frac{2}{x^2 + 4}$

(a) Find the intervals where the function is increasing/decreasing and the coordinates of all local max/min.

(b) Find the intervals where the function is concave up/down and the coordinates at all inflection points.

(c) Find the limiting behaviour as x approaches $\pm\infty$.

(d) Use the above information to sketch a rough graph of the function. Include all intercepts on graph.

9. Let $f(x) = \frac{4-x}{3+x}$.

- (a) Find the intercepts of f and all its asymptotes.

Find the limiting behaviour near each vertical asymptotes ($\lim_{x \rightarrow a^\pm} f(x)$).

Use this information to sketch a guess for the graph of f .

- (b) Confirm details of your guess by finding the intervals where the function is increasing/decreasing and concave up/down. Clearly mark any local extrema and inflection points in your plot.

- (c) Use algebra (long division in this case) to rewrite the function as a polynomial plus a proper quotient. Could you have obtained the graph of f from this result?

10. Consider the function

$$f(x) = \frac{\sin x}{x}$$

(a) State the domain and all intercepts of the function.

(b) Show that f is even.

(c) Find all asymptotes of the function.

(d) Explain why the following limit equals the derivative of $g(x) = \sin x$ at $x = 0$ and use that to deduce that

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

(e) Sketch a graph of the function that reflects all the information you found above.

11. The reaction rate V of a common enzyme reaction is given in terms of substrate level S by

$$V = \frac{V_o S}{K + S}, \quad S \geq 0$$

where V_o and K are positive constants.

- (a) Show that V is an increasing function of S . Explain why it follows that V has no absolute maximum value.

- (b) What is $\lim_{S \rightarrow \infty} V$?

- (c) Determine the concavity of the graph of V (where is it concave up? where concave down?).

- (d) Sketch a graph of V as a function of S that reflects all the information you found above.